

Literature Review

Visual navigation for UAVs

Occam's razor: When is complex navigation needed?

A question for the research agenda: could a planned route at a suitable cruise altitude handle much of ordinary delivery?

SIMPLE BASELINE

Climb, cruise, approach

[Prime Air](#) publishes planned routes and cruise altitudes of 205–250 ft outbound and 325–370 ft returning; it descends near the delivery point.

A street-level route must show a mission benefit over this option.

TASK REQUIRES PROXIMITY

Reach an obscured surface

Under-bridge inspection requires views that are unavailable from above. GNSS and obstacle clearance can also become difficult under the structure.

[Inspection operator account](#): evidence for the task need, not a benchmark for a reviewed paper.

Open question: Which missions defeat the simpler flight design, and does added navigation improve completion, interventions, time, or energy? The examples above motivate the question; they do not settle it for every mission.

Motivation: What visual navigation requires

A navigation result depends on the goal, what the UAV knows about its position and surroundings, and whether it can execute a safe route.

Interpret the goal

A waypoint, image, object, or route instruction leaves different decisions to the UAV.

[AerialVLN](#) studies route instructions; [AirHunt](#) studies object search.

Keep track of the scene

Estimate position, remember landmarks, and distinguish observed space from guesses.

[AeroBelief](#) maintains target beliefs; [VISTA](#) studies active mapping for exploration.

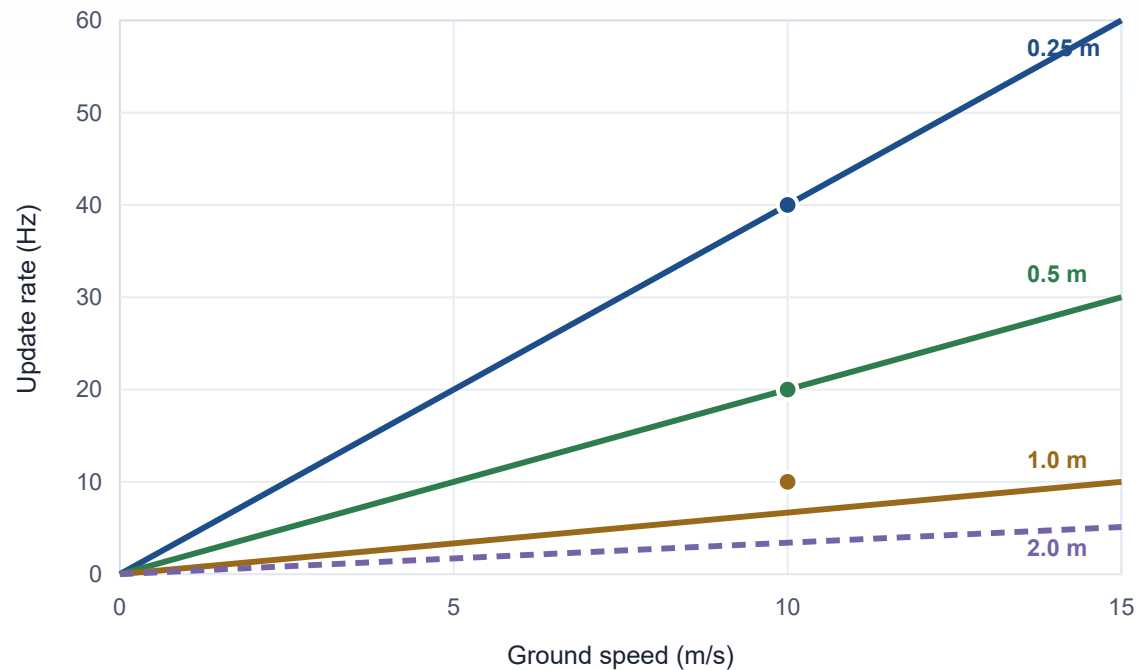
Move safely in time

Choose a feasible path and act before new obstacles or stale observations make it unsafe.

[FlowPilot](#) predicts trajectories; [LiteVLA-H](#) separates fast and slow updates.

Motivation: Flight speed sets the update rate

If no more than **s** metres should pass between usable updates, the minimum update rate is **f = v / s**.



AT 10 M/S

1.0 m per update	10 Hz
0.5 m per update	20 Hz
0.25 m per update	40 Hz

These are perception or planning updates. At 5 m/s, a 0.4 s image-to-command delay means **2 m** travelled; at 10 m/s, **4 m**.

Environments: Where systems train and where they are tested

STUDY	TRAINING ENVIRONMENT / DATA	TEST ENVIRONMENT
AerialVLN	Synthetic outdoor city scenes and instruction trajectories.	Held-out routes in its flight simulator; no physical flight.
CityNav	Human trajectories in CityFlight, built from scans of Cambridge and Birmingham.	Held-out routes in CityFlight; the real-city scans are still a simulator.
OpenFly	Rendered aerial trajectories from Unreal, GTA V, Google Earth, and 3DGS scenes.	Benchmark splits in rendered scenes; limited outdoor flight is reported separately.
SOUS VIDE	FiGS: reconstructed Gaussian scene plus drone dynamics; expert trajectories distilled to a policy.	105 physical-flight trials, including mass, wind, lighting, and scene changes.
AirDreamer	World-model and policy learning in simulation.	Unseen simulated environments and real drone flights without retuning.
GRaD-Nav++	Differentiable drone dynamics in a photorealistic 3DGS simulator.	New simulated settings and real quadrotor flight.

Goals: What the UAV is asked to reach

INPUT	WHAT IT GIVES THE UAV	WHAT STILL HAS TO BE SOLVED	EXAMPLES
Waypoint or pose	A target position in a stated coordinate frame.	Localization, frame conversion, route choice, obstacle clearance, and stopping.	Common in flight tasks; state the frame and map available.
Destination image	A view of the target place or instance.	Recognize it from a new altitude or heading; search and remember visited places.	Less often isolated in the aerial papers reviewed here.
Object or category	A phrase such as "find the water tower."	Search, distinguish candidates, estimate location, and confirm a match.	AirHunt , AeroBelief , AECNav
Route instruction	Landmarks, turns, and their order.	Ground each phrase in the view, retain subgoals, and recover after a missed landmark.	AerialVLN , OpenUAV , FlightGPT

Current implementations: VLMs and VLAs

Related work: [VL-JEPA](#) learns visual-language answer features; [OpenVLA](#) produces robot actions. Neither tests UAV flight.

PAPER	USE ON THE UAV	REPORTED EVIDENCE	LIMIT IN REVIEWED EVIDENCE
FlightGPT	VLM selects actions from images and route instructions.	Improved unseen-route success on CityNav.	Response time unreported; no explicit staged planner.
VLM-Nav	VLM labels obstacles; range sensing helps select five actions.	98% completion in two AirSim settings.	AirSim only; no action-consequence model.
AirHunt	VLM object semantics guide continuous aerial search.	Improved object-search planning.	Target identity after occlusion is untested.
SkyVLN	VLM plans a path; nonlinear MPC controls flight.	AirSim / Unreal urban simulation.	No physical-flight or response-time result in the paper.
GRaD-Nav++	VLA trained in a differentiable Gaussian flight simulator.	Onboard hardware evaluation.	Calibrated uncertainty and delay handling unshown.

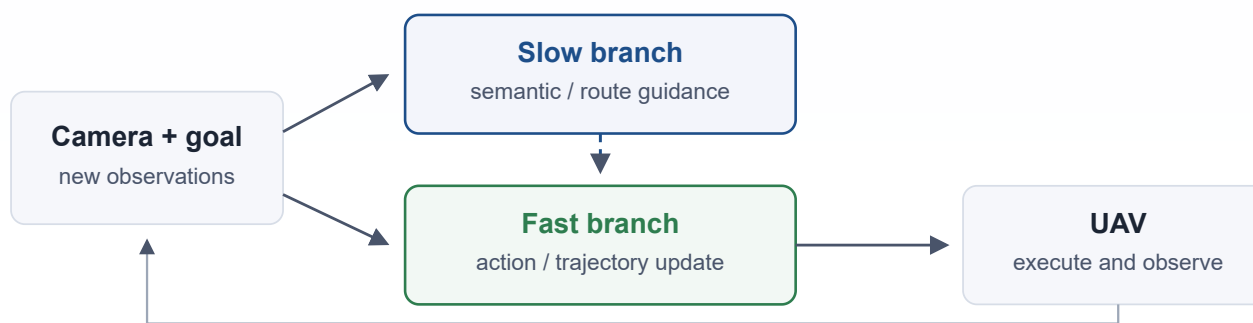
Current implementations: Aerial world models

Related work: [DreamerV3](#) learns from imagined rollouts; [DINO-WM](#) plans with predicted visual features. Aerial deployment requires its own test.

PAPER	PREDICTION AND UAV USE	REPORTED EVIDENCE	LIMIT IN REVIEWED EVIDENCE
SkyJEPA	Predicts quadrotor state under candidate actions for control.	Long-horizon sim-to-real control.	State input; no camera or language task.
AirDreamer	Latent futures train a drone navigation policy.	Real flights up to 1.8 m/s without retuning.	Calibrated risk and safety semantics unshown.
FlowPilot	Predicts future depth and an executable trajectory.	Real-time quadrotor deployment.	No language goal or end-to-end action-age result.
WorldFly	Predicts future camera view with language-conditioned action.	87% seen; 31% unseen benchmark success.	Future-frame compute; real flight unshown.
ImagineUAV	World-action prediction feeds a kinodynamic planner.	Aerial planning method in the source review.	Our notes describe the method, but do not audit real-flight evaluation.

Current implementations: Fast and slow flight loops

These systems separate slower semantic decisions from frequent flight updates, but place the boundary at different points.



PAPER	SLOW SIDE	FAST SIDE	EVIDENCE / LIMIT
FSD-VLN	VLM semantic priors.	Diffusion action stream.	Long-route simulation gains; applied action age unreported.
LiteVLA-H	Periodic semantic perception.	Action-token guidance: 50.65 ms onboard model step.	Full sensor-to-actuator delay unreported.
VLM-MPPI	VLM selects a trajectory.	MPPI replans at 20 Hz.	Real quadrotor tests; 20 Hz is not VLM response rate.

Current implementations: Architectural designs for memory

PAPER	DESIGN	REPORTED EVIDENCE	LIMIT IN REVIEWED EVIDENCE
VISTA	Online semantic Gaussian map for exploration.	Real-quadrotor exploration.	Map is not a predictive or certified safety state.
ATLAS Navigator	Language-embedded map selects task-relevant views.	Real ground-robot exploration.	Transfer to UAV flight and its safety constraints unshown.
RAVEN	Persistent semantic-spatial memory for search.	Outdoor aerial object search.	Our review does not establish multi-stage route recovery.
AeroBelief	Separate semantic and spatial target beliefs.	Aerial object-goal navigation.	Calibration and occlusion recovery unshown.
AECNav	Chooses the next view to collect target evidence.	Open-vocabulary object navigation.	Identity persistence through missed detections unshown.

Research gaps: Real-world validation

City-scale mapless route following has not been shown on a real UAV in this review.

CURRENT EVIDENCE

[SPRIN-D](#) reports a 1,371 m real sortie and a 1,023 m urban sortie using onboard sensors and prior geodata. Its 9 km course was the challenge target. [CityNav](#) uses real-city scans, but routes run in simulation.

NEEDED

Fly held-out city routes with the same goal input and stated map access. Report arrival, collisions, minimum clearance, human intervention, time, and energy. Compare with planned overflight when the mission permits it.

What use do the papers establish?

RESEARCH	PLAUSIBLE JOB	EVIDENCE BOUNDARY
FlowPilot / AirDreamer	Local flight around clutter during an approach or inspection.	Real flights are short; they do not justify city-wide weaving for ordinary delivery.
SPRIN-D	Keep position estimates useful on long GNSS-denied waypoint missions.	Kilometer sorties use LiDAR and prior geographic data; not arbitrary city autonomy.
CityNav / aerial VLN	Study grounding of language-described destinations in city scenes.	Language may be unnecessary when coordinates are available; benchmark success is simulated.
World models	Potentially improve a decision under uncertainty.	Prediction quality matters only if it improves flight outcomes against a simpler planner.

Application evidence and scientific evidence differ. A paper can isolate a useful capability without proving that a complete commercial mission needs its full stack.

Research gaps: Long-horizon capability

Long routes are studied, but missed landmarks and stage-by-stage recovery remain weakly tested.

CURRENT EVIDENCE

FSD-VLN reports gains on long aerial instruction routes. **RAVEN** keeps spatial memory for outdoor object search. These test different goals.

NEEDED

Test multi-stage instructions when a landmark is missed or obscured. Report subgoal order, wrong turns, recovery distance, and final arrival on held-out routes.

Research gaps: Onboard computation

A model rate alone cannot tell us how far the UAV flies before a command applies.

CURRENT EVIDENCE

LiteVLA-H reports a 50.65 ms fast model step, but our notes do not pair it with flight speed or full image-to-action delay. VLM-MPPI's 20 Hz is its planner rate, not VLM latency.

NEEDED

Illustration: at 5 m/s, a 0.4 s image-to-applied-action delay means 2 m of travel. Report both quantities in the same flight, plus late commands, clearance, and onboard power.

Research gaps: Dynamic obstacles

Moving obstacles can invalidate a path while the system is perceiving and planning.

CURRENT EVIDENCE

[ImagineUAV](#) couples learned future prediction to a feasible-path planner. [FastBridge](#) adds a safety filter whose guarantees depend on its map and model assumptions.

NEEDED

Cross moving obstacles at varied speeds and observation delays. Measure collision rate, minimum clearance, and whether the UAV replans when a prediction is wrong.

Research gaps: Sim-to-real robustness

Real-flight transfer is demonstrated, but the studies do not give a matched robustness comparison.

CURRENT EVIDENCE

[SOUS VIDE / FiGS](#) trains a flight policy in a Gaussian-scene simulator and tests real flight under changes in mass, wind, and lighting. [AirDreamer](#) flies a world-model policy at up to 1.8 m/s without retuning.

NEEDED

These are different flight tasks and tests. Hold the UAV, sensors, goal, and controller fixed; compare simulation with real flight under held-out wind, lighting, and sensor degradation. Report success and clearance.

References: Aerial tasks and policies

BENCHMARKS AND ROUTE TASKS

AerialVLN - aerial vision-language navigation
CityNav - real-city scenes for aerial navigation
OpenFly - aerial VLN platform and dataset
OpenUAV - open-ended UAV object finding

LANGUAGE AND FLIGHT ACTIONS

FlightGPT - UAV vision-language navigation
SkyVLN - VLM path planning with nonlinear MPC
AeroVLA / AerialVLA - aerial vision-language-action
FSD-VLN - fast-slow aerial VLN
LiteVLA-H - dual-rate onboard guidance
VLM-MPPI - VLM selection over fast trajectories

AERIAL SEARCH AND TRACKING

AirHunt - object search and continuous planning
AeroBelief - semantic-spatial target belief
AECNav - active evidence consolidation
RAVEN - persistent spatial memory
Fast-Tracker 2.0 - active aerial tracking
OA-VAT - instance tracking and recovery

VLM NAVIGATION

VLM-Nav, "Mapless UAV navigation using monocular vision driven by vision-language models," PLOS ONE, 2026. See the supplementary review and local paper notes for its publication record.

References: Prediction and scene models

AERIAL WORLD MODELS

SkyJEPA - quadrotor state/action prediction
FlowPilot - future depth and trajectory
WorldFly - world-model VLA for UAV navigation
AirDreamer - drone navigation with world models
ImagineUAV - world-action model and kinodynamic planning

MAPS AND EXPLORATION

FUEL - frontier-based UAV exploration
VISTA - semantic Gaussian exploration on a quadrotor
ATLAS Navigator - language maps on a ground robot
SCOUT - semantic uncertainty on a ground robot
SOUS VIDE / FiGS - Gaussian-scene sim-to-real
GRaD-Nav++ - differentiable simulator and aerial VLA

GEOMETRY AND SAFETY

SemSafe-3DGS - semantic risk in Gaussian maps
FastBridge - safety filters over Gaussian scenes
ASMA - scene-aware barrier functions
Certifiably-Correct Mapping - mapping with stated assumptions

COMPANION SOURCE NOTES

Supplementary literature review
Presentation notes
UAV papers in depth
Historical VLM source notes